

# CAVE SHIPPERS CO. LLC SUPPLY CHAIN AND SALES ANALYTICS **PROJECT**

Business Intelligence Case Study using MySQL,  
R, and Tableau



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# OVERVIEW

A mock dataset was created from a similarly aligned company of shipment data where the dataset's size and metadata were assessed to understand structure and determine the appropriate handling approach based on each field's attributes.

- 01 Overview
- 02 Business Problem
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- 07 Customer & Quantity Insights
- 08 Statistical Analysis & Outliers
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# Business Problem



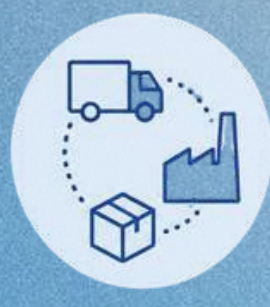
Sales  
Growth



Forecasting



Customer  
Trends



Supply Chain  
Analytics

## Business Challenges

Cave LLC sought to better understand:

- top-performing container materials,
- customer purchasing trends,
- quarterly sales behavior,
- and future sales growth opportunities.







# OBJECTIVES

**01**

Clean and standardize raw sales data

**02**

Analyze material profitability

**03**

Identify customer purchasing trends

**04**

Detect sales anomalies/outliers

**05**

Forecast future sales growth

**06**

Deliver actionable business insights



# Data Architecture





# SALES PERFORMANCE ANALYSIS

## Material Sales Performance

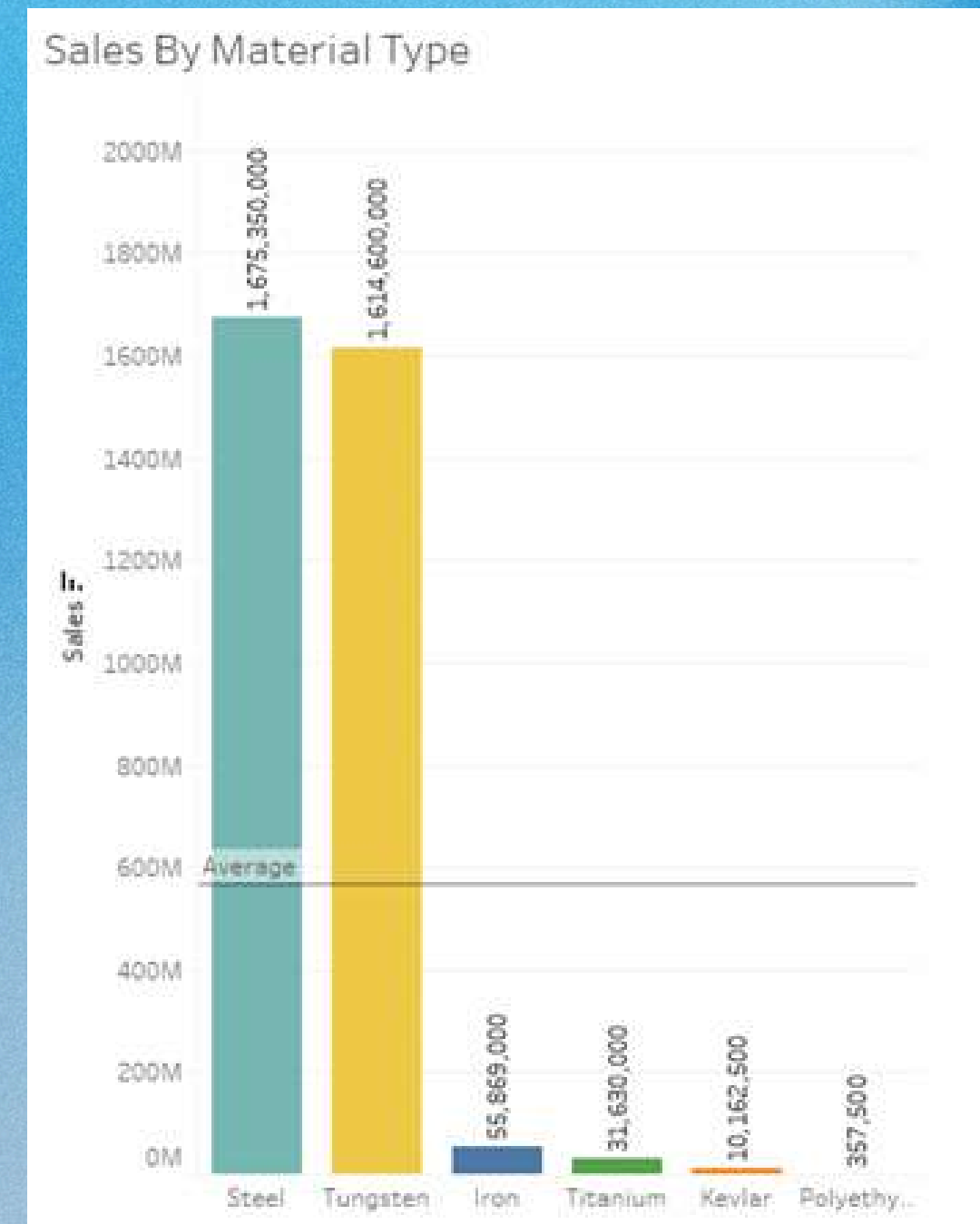


### Key Findings

- Steel and Tungsten generated the highest sales
- Polyethylene showed the weakest demand
- Sales concentration favored traditional industrial materials



Traditional container materials continue to dominate customer preference and revenue generation.





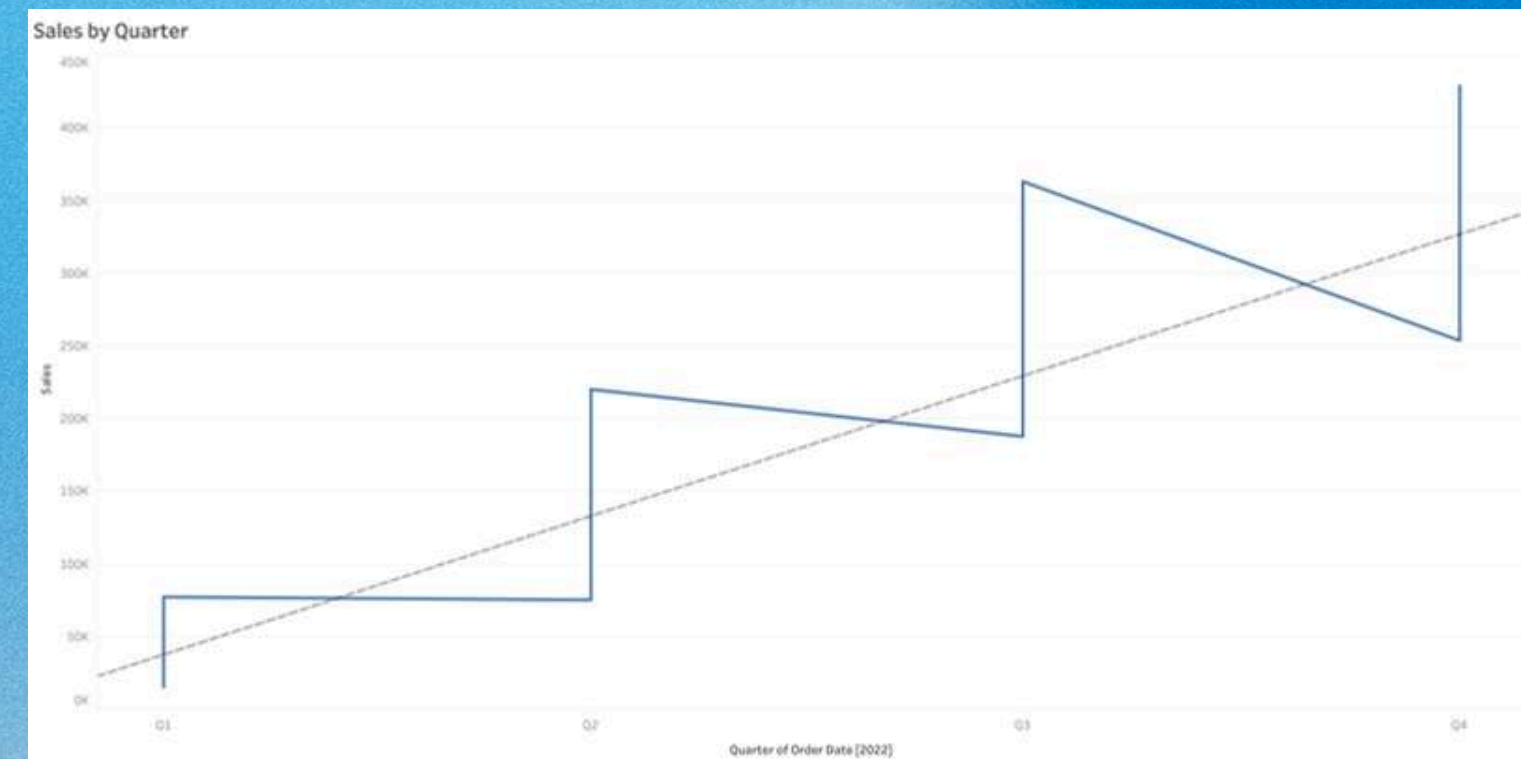
# QUARTERLY SALES TRENDS

## Material Sales Performance



### Key Findings

- Q2 and Q3 showed strongest growth momentum
- Q3 demonstrated highest sales volatility
- Q4 contained limited data but indicated continued upward trajectory





# CUSTOMER AND QUANTITY INSIGHTS

## Material Sales Performance



### Business Insights

- Higher unit purchases correlated with pricing discounts
- Customers appeared to purchase heavily at beginning of quarters
- Bulk purchasing impacted overall sales margins





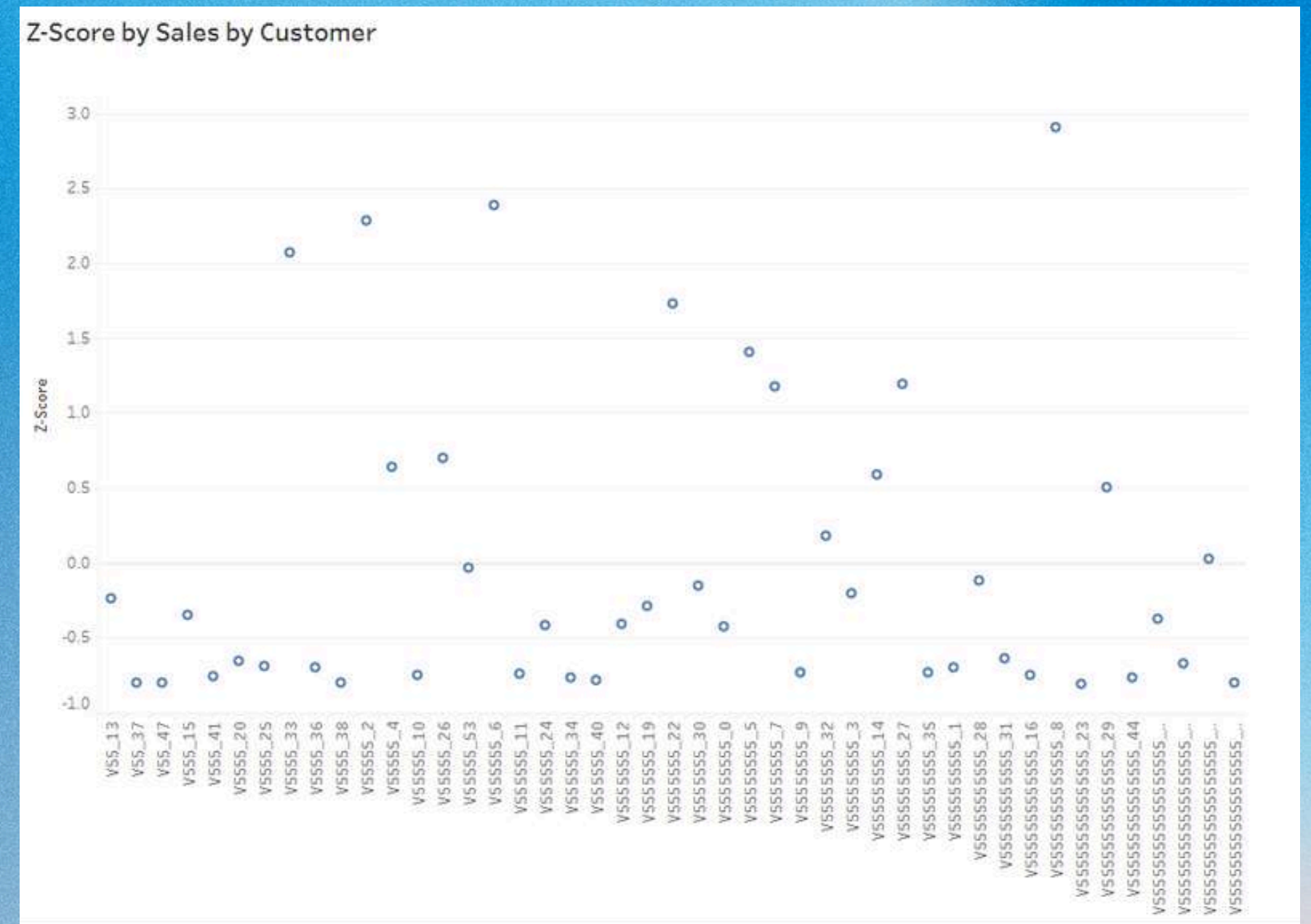
# STATISTICAL ANALYSIS AND OUTLIERS

Z-score analysis was used to identify unusual sales patterns and purchasing anomalies.



## Findings

- Steel sales behaved as statistical outliers due to unusually high demand
- One order exceeded normal frequency thresholds during Q4
- Customer sales activity remained generally stable





# FORECASTING & PREDICTIVE ANALYSIS

## 10-Year Sales Forecast

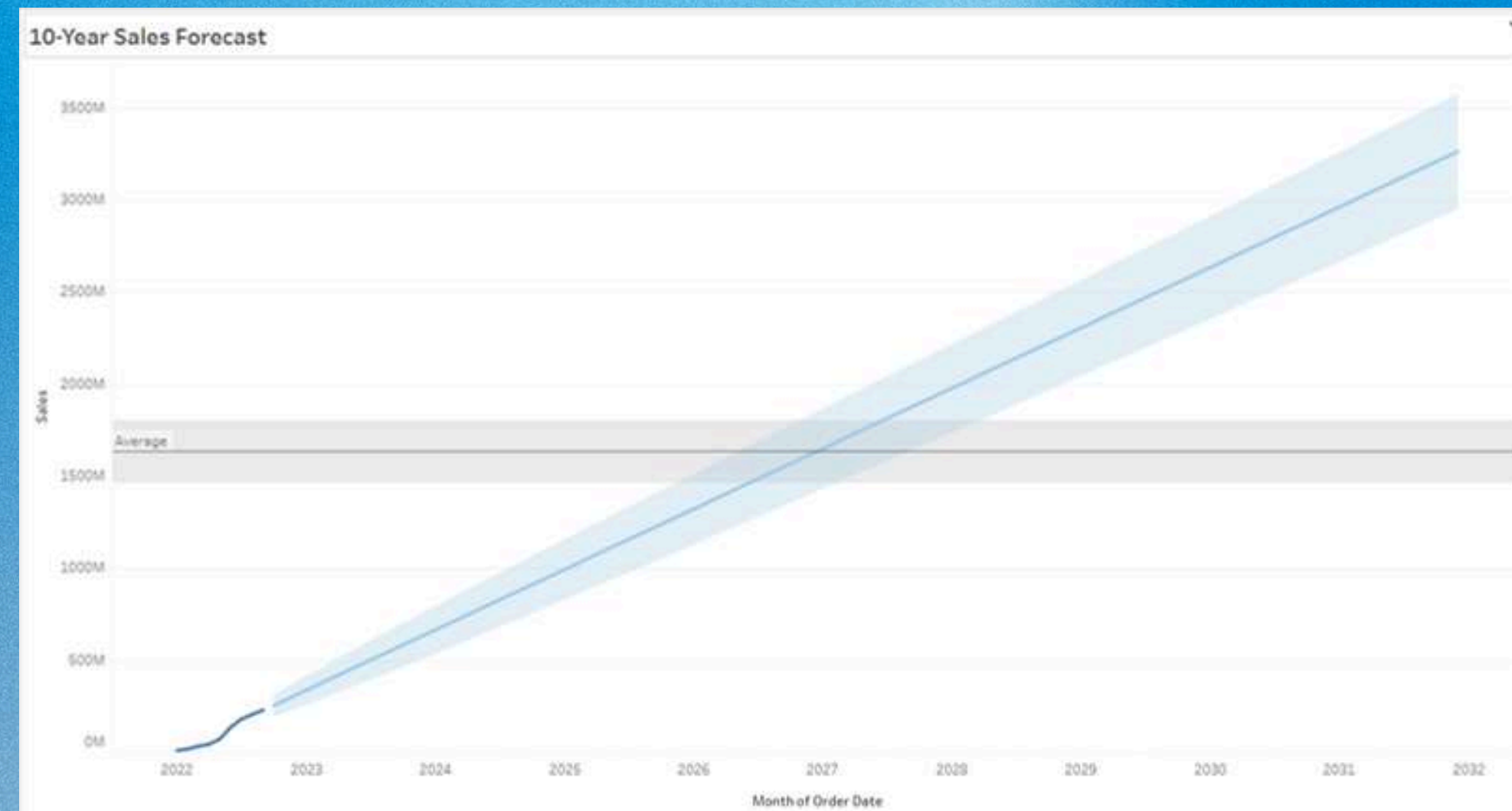


### Key Findings

- Forecast models projected continued upward sales growth
- Titanium and Tungsten showed strongest long-term growth potential
- Iron and Kevlar displayed inconsistent forecast behavior



Forecast reliability was limited by short historical time range.





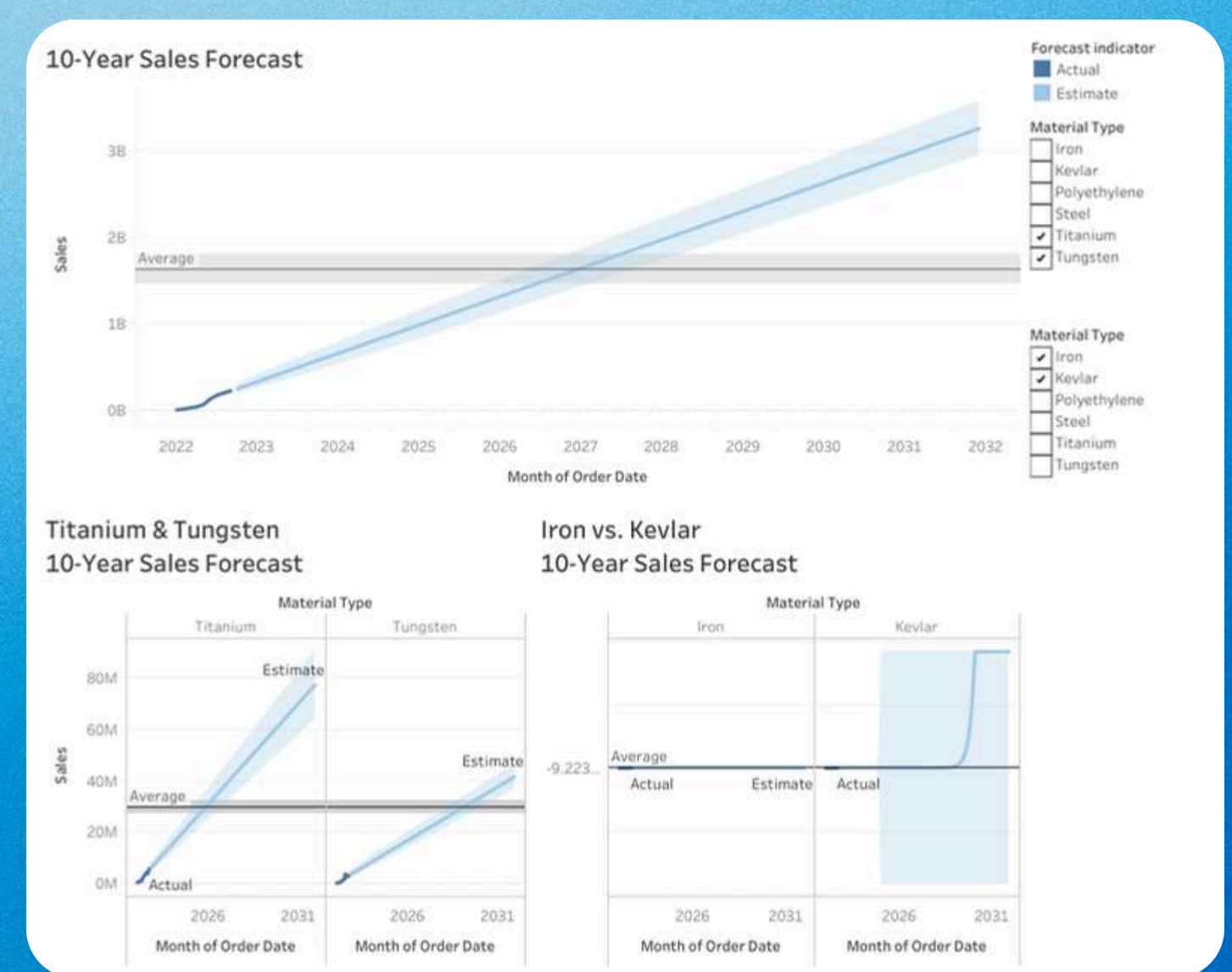
# DASHBOARD SHOWCASE



Sales Dashboard

## Dashboard Features

- Sales by quarter
- Material performance
- Forecast visualization
- Customer trend analysis
- Statistical anomaly tracking



Forecast Dashboard





# Final Recommendations

- Continue prioritizing Steel and Tungsten production
- Investigate Q4 sales anomalies for operational insights
- Review discount structures impacting bulk purchases
- Expand forecasting with additional historical data
- Implement automated BI dashboards for leadership reporting



# Business Impact

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How data-driven decisions translate into measurable outcomes across the organization.

Transform raw data into actionable insights

Perform statistical analysis

Build business intelligence dashboards

Communicate findings to stakeholders